Problem 1. *Let* $\vec{a} = [3 \quad 4]$ *and* $\vec{b} = [1 \quad -2]$
Write both vectors in column vector notation and find their magnitudes

Problem 2. *Let* $\vec{a} = [5 \quad 0 \quad -6 \quad 8]$ *and* $\vec{b} = [11 \quad -12 \quad -4 \quad 7]$
Write both vectors in column vector notation and find their magnitudes.
Show that $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$

Problem 3. *Let* $\vec{u} = [2 \quad -1 \quad 4]$ *and* $\vec{v} = [-3 \quad 0 \quad 5]$
Find

1. $\vec{u} + \vec{v}$
2. $2\vec{u} - \vec{v}$

Problem 4. *Let* $\vec{x} = [8 \quad -4 \quad 2 \quad 5]$, $\vec{y} = [-4 \quad 4 \quad 0 \quad 7]$ *and* $\vec{z} = [2 \quad 3 \quad -2 \quad 1]$
Find

$$\vec{x} + 3\vec{y} - 2\vec{z}$$

Problem 5. *Let* $\vec{a} = [1 \quad 2]$ *and* $\vec{b} = [2 \quad 3]$
Find the angle θ *between them using the dot product*

Problem 6. *Given vector* $\vec{v} = [6 \quad -8]$, *find a unit vector in the same direction.*

Problem 7. *Let* $\vec{a} = [3 \quad 4]$ *and* $\vec{b} = [1 \quad 0]$. *Find the projection of* $\vec{a}$ *onto* $\vec{b}$. *Find the projection of* $\vec{b}$ *onto* $\vec{a}$

Problem 8. *Let* $\vec{p} = [2 \quad -1, \quad 3]$ *and* $\vec{q} = [0 \quad 4, \quad -1]$. *Verify*

$$\vec{p} \cdot \vec{q} = \vec{q} \cdot \vec{p}$$

$$\vec{p} \cdot (\vec{p} + \vec{q}) = \vec{p} \cdot \vec{p} + \vec{p} \cdot \vec{q}$$

Problem 9. *Determine whether the vectors* $\vec{a} = [1 \quad 2]$, $\vec{b} = [2 \quad 4]$ *and* $\vec{c} = [-1 \quad -2]$ *are linearly independent.*

Problem 10. *Are the vectors* $\vec{u} = [3 \quad 6]$ *and* $\vec{v} = [-2 \quad -4]$ *collinear?*

Problem 11. *Bonus Problem I*
Let $\vec{a} = [1 \quad 0 \quad 0]$ *and* $\vec{b} = [0 \quad 1 \quad 0]$. *What is the angle between* $\vec{a}$ *and* $\vec{b}$?
Find the vector cross product $\vec{a} \times \vec{b}$. *Show that the resultant vector is in a direction normal to both* $\vec{a}$ *and* $\vec{b}$

Problem 12. *Bonus Problem II*
Let $\vec{a} = [4 \quad a]$ *and* $\vec{b} = [2 \quad 3]$. *Find the area of the parallelogram spanned by* $\vec{a}$ *and* $\vec{b}$.